Wadim **Müller**

Embedded Software Expert · Linux Enthusiast

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"He who stops being better stops being good." (Oliver Cromwell)

QAVION Recruitment Team

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QUAVION GROUP GmbH

Job Application for Senior Embedded Engineer

Dear Ms. Karimi,

I am writing to express my enthusiastic interest in the Senior Embedded Engineer position I found on linkedin.com. As an embedded software engineer with over a decade of experience solving technically challenging problems across diverse CPU/SoC platforms, and with direct contributions to the mainline Linux Kernel, I am confident that my skills are an excellent match for the requirements of this role.

In my current position at the startup CMBlu, a startup focused on energy storage solutions, I am developing a Battery Management System based on Yocto Linux. My responsibilities include writing and integrating Linux Kernel drivers for our custom boards and leading the board bring-up of our mainboard, which is based on the TI AM64xx SoC. This role has further sharpened my skills in low-level driver development and system integration.

Previously, at Continental, I was responsible for integrating SoC-vendor Board-Support-Packages into Greenhills INTEGRITY OS-powered products. A significant achievement was the creation of a full machine emulation for Renesas RCar3 SoC-based boards using QEMU. This required developing emulations for numerous SoC-specific peripherals and the bootrom to successfully boot a proprietary OS from the reset vector, work which can be viewed on my QEMU GitHub fork.

In addition, I developed a comprehensive QEMU emulation of the NXP S32G Cortex-M7 tri-core system, including custom device models for Ethernet and CAN controllers. This setup enabled the unmodified Classic AUTOSAR stack—used in a Continental SDV platform—to boot and run successfully within QEMU.

My experience also includes developing a PCI Endpoint and the necessary drivers to pass through block devices between a PCI host and endpoint on the NXP S32G SoC. This involved writing both an endpoint and a PCI device driver for the Linux kernel, and I have posted the patch on the Linux Kernel Mailing List, which can be viewed here: lwn.net/Articles/963538/.

I am also passionate about driving process improvements. To aid in failure analysis for a Classic AUTOSAR project, I developed a unique Call Stack Analysis Tool capable of determining the worst-case stack usage by analyzing machine instructions directly from an ELF image. Its most innovative feature is the ability to detect calls made through function pointers by examining object file relocation entries.

My Master's Thesis further expanded my expertise, involving the development of an FPGA-accelerated Linux system for real-time video feature extraction on a Xilinx Zynq PSoC. This project included designing two IP-Cores in Verilog and developing the corresponding Linux kernel device drivers. The IP-Core designs, kernel drivers, device tree, and demo application are all available on my GitHub profile.

I would also like to highlight earlier career achievements: At Rohde & Schwarz, I ported FreeRTOS to an SH2A embedded CPU using GCC. At Magna Powertrain, I ported FreeRTOS to the NXP S12Z platform using the CodeWarrior toolchain.

Thank you for considering my application. I would welcome the opportunity to bring my embedded systems experience, Linux kernel expertise, and passion for solving hard technical problems to your team.

Sincerely,

Wadim Müller